

CRICKET PLAYER PERFORMANCE TRACKING APPLICATION

1. BRAINSTORMING AND IDEATION PHASE

1.1 Introduction and Problem Identification

Cricket performance is shaped by many connected activities: batting, bowling, fielding, match situations, playing conditions and team strategy. A player's contribution cannot always be understood from a single score or a match result. Coaches and players need records that show performance across several matches and that make it possible to review progress over a season. In many schools, colleges, academies and local clubs, these records are maintained in paper scorebooks or separate spreadsheets. Although these methods are familiar, they can make long-term tracking time-consuming and inconsistent.

Manual records may contain spelling variations, missing values, calculation mistakes or different formats for the same statistic. For example, one scorer may record bowling figures using cricket overs notation while another may enter decimal values. If the values are interpreted incorrectly, economy rate calculations can become inaccurate. Searching for a player's previous match figures may also require checking multiple documents. When a coach wants to compare performances across a selected period, the information may need to be copied and calculated again.

The proposed Cricket Player Performance Tracking Application addresses these issues by organizing player profiles, teams, matches and scorecard statistics in a centralized digital system. The application is intended to reduce repetitive work, standardize data entry and provide automatically calculated summaries. It supports performance review by presenting match records and selected trends in a clear format. The system is a decision-support tool: it helps users examine available statistics, while final coaching, training and team-selection decisions remain with responsible people.

The problem identification stage therefore establishes three core needs: reliable storage of cricket records, consistent calculation of performance indicators, and convenient retrieval of historical information. These needs guide the project's scope and acceptance criteria. The project should not claim to predict a player's future performance unless a separate predictive model is developed and validated. Its primary purpose is accurate tracking and understandable reporting of recorded performance.

1.2 Idea Generation and Proposed Solution

Idea generation considered several ways to manage cricket statistics. Paper scorebooks are inexpensive and useful during a match, but their information is difficult to search and aggregate. Spreadsheets support formulas and charts, yet separate files can become inconsistent, and formulas may be accidentally changed. General-purpose scoring applications may offer live scoring but may not provide the specific historical reports or player-management workflow required by a college or academy project.

The proposed solution is a focused application that stores player and match information in a structured database. Users can register players, create teams, enter match details and record batting, bowling and fielding contributions. The application then calculates selected metrics and displays them through tables, summary cards and charts. A report can be filtered by player, team, season, date range or match format, depending on the implemented features. Export to CSV allows the records to be used in spreadsheet software for further review.

The solution should be developed in stages. The minimum viable version includes player management, match registration, performance entry, validated calculations and a basic report. Additional features can include player comparison, season filters, downloadable reports, role-based access and data import. Live scoring, video analysis, wearable-device integration and automated performance prediction are outside the initial scope unless additional data sources and testing are available.

The idea is considered useful because it combines structured record keeping with automatic calculations while remaining practical for an academic implementation. The design should prioritize accuracy and ease of use over unnecessary complexity. Before implementation, sample scorecards and calculation definitions should be reviewed with a knowledgeable scorer or coach. This reduces the risk of building a technically functional application that interprets cricket statistics incorrectly.

1.3 Stakeholders, Users and Use Cases

The main stakeholders are players, coaches, scorers, team administrators and the project administrator. Each stakeholder interacts with the application for a different purpose. Players want to review personal statistics and match history. Coaches want to examine player contributions, compare records within a defined period and identify areas for training.

Scorers need forms that are quick to use and that prevent common entry mistakes. Administrators maintain teams, players and access permissions.

Typical use cases begin with player registration. An authorized user enters a player's name and relevant playing attributes, such as batting style, bowling style and role. A team can then be created and players associated with it. When a match is scheduled or completed, the user records the date, venue, format and participating teams. Performance details are entered for each innings, and the system validates the values before saving them.

After data entry, users can view a player profile, inspect a match scorecard, filter a performance history or generate a summary. A coach may compare two players' batting strike rates over the same set of matches, while a player may review personal runs scored across a season. Such comparisons must clearly identify the selected period and number of matches so that users do not confuse a small sample with a complete record.

Use cases should specify the actor, starting conditions, main steps, expected outcome and exception handling. For example, saving a batting record requires a valid player and innings, non-negative runs and balls faced, and a consistent dismissal status. If required information is missing, the system should explain what needs correction and preserve the other entered values. Defining use cases early helps the development team build workflows around real user tasks rather than isolated screens.

1.4 Scope, Feasibility and Initial Success Criteria

The project scope includes player and team management, match registration, batting and bowling performance entry, selected fielding records, automated statistics, search, dashboards and report export where supported. The system is intended for a defined group such as a college team, academy or local club. It assumes that match data is entered by an authorized user and checked against a scorecard. The first version does not depend on an external live-scoring service.

Technical feasibility depends on the chosen programming language, user-interface framework and database. A Python-based application with a lightweight interface and SQLite database is suitable for a small prototype. A larger multi-user deployment may require a server database, authentication and backup procedures. Operational feasibility depends on whether scorers can use the forms with minimal training and whether coaches understand the displayed

metrics. Economic feasibility is supported by open-source tools, although deployment, maintenance and training still require time.

The project should define measurable success criteria. Core records must be saved and retrieved correctly. Calculations must match verified sample scorecards. Invalid values such as negative runs should be rejected with understandable messages. The system should distinguish zero from missing or unavailable data. Reports should apply filters correctly and state their scope. A new user should be able to complete common tasks without extensive assistance.

Initial feasibility review should also identify constraints such as limited development time, small test datasets, device availability and the need to protect player information. These constraints influence which features belong in the first release. A clear scope prevents uncontrolled expansion and gives the team a practical basis for planning, implementation and final evaluation.